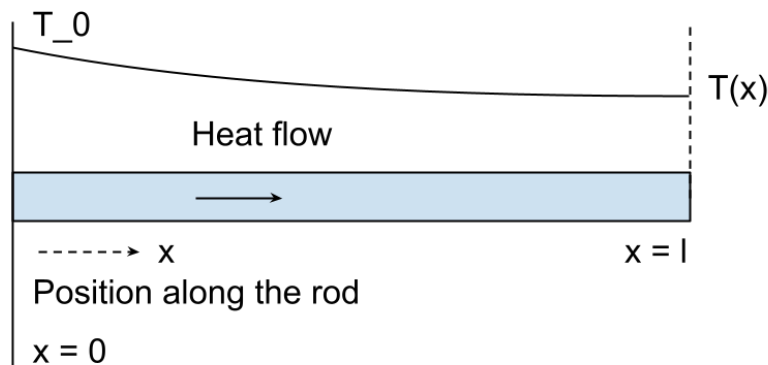


Thermal Conduction

- Introduction
- Temperature
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- Heat Transfer with a Small Segment
- Why Exponentials Appear in the Solution
- Hyperbolic Functions
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Introduction

Analysis of temperature distribution along a rod using Fourier's law in analytical methods, electrical analogies and finite element methods.



Temperature

Temperature is a measure of the kinetic energy of particles in a material:

- Atoms and molecules in a solid or fluid move or vibrate
- The faster they move the higher the temperature
- Temperature differences drive heat flow from hot regions to cold regions

In this modelling $\theta(x)$ is the temperature difference relative to the surroundings representing the energy available to flow along or out of the rod:

$$\theta(x) = T(x) - T_{\infty}$$

- $T(x)$ - actual temperature at position x along the rod
- T_{∞} - temperature of the surroundings
- $\theta(x)$ - temperature difference

Law of Heat Conduction

The law describing how heat moves through a solid was discovered experimentally by French physicist Joseph Fourier in 1822. Fourier observed that heat always flows from regions of higher temperature toward regions of lower temperature.

Fourier's Law is the fundamental law of heat conduction:

$$q = -kA \frac{dT}{dx}$$

- q - heat transfer rate
- k - thermal conductivity of the material
- A - cross-sectional area through which heat flows
- $\frac{dT}{dx}$ - temperature gradient along the material

A large temperature gradient means temperature is changing rapidly along the material, which drives a larger heat flow. While the negative sign produces a positive heat flow, indicating that heat moves in the positive x direction in our example. Materials with higher thermal conductivity transfer heat more easily.

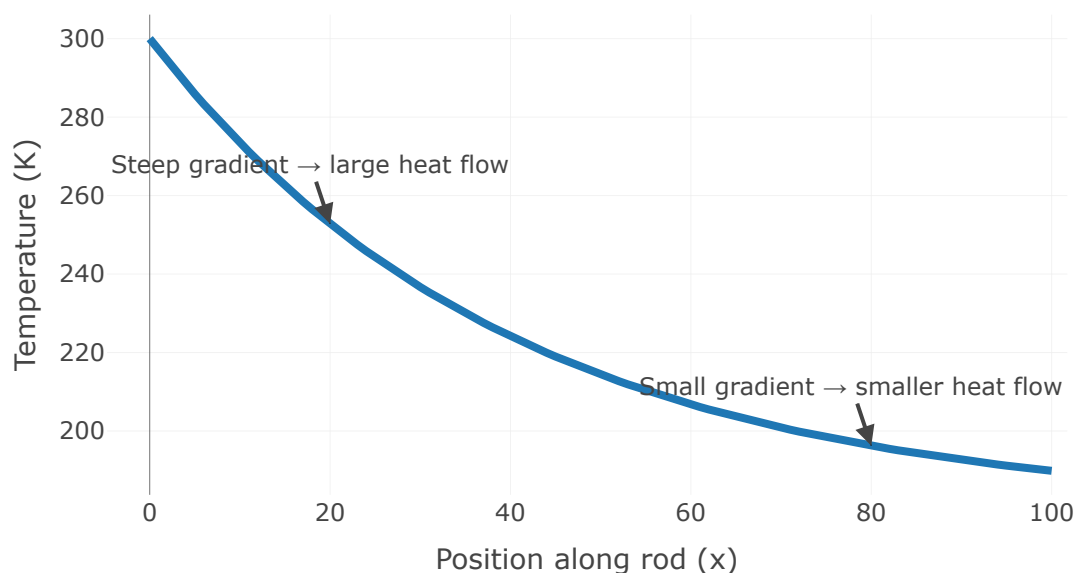
For our cylindrical rod, heat entering a small segment at position x can be written as:

$$q_x = -kA \frac{dT}{dx}$$

At the other side of the segment, $x + dx$, the heat flow will be slightly different because the temperature gradient has changed. The heat leaving the segment can be written as:

$$q_{x+dx} = -kA \frac{dT}{dx} \Big|_{x+dx}$$

The difference between these two heat flows represents the amount of heat that is lost or gained in this segment.



Heat Transfer with a Small Segment

Now we combine the ideas of energy conservation and Fourier's law on a very small segment, which is termed a "control volume".

In addition to axial conduction, the segment can transfer heat radially with the surroundings. We call this radial transfer q_r .

By convection:

$$q_r = hP(T - T_\infty)dx$$

- h - convective heat transfer coefficient of the surrounding layer
- P - perimeter of the rod
- T_∞ - temperature of the surroundings
- dx - length of the segment

Applying the energy balance to the segment at steady state (no accumulation, no internal generation):

$$\text{Energy In} - \text{Energy Out} - \text{Radial Transfer} = 0$$

Substituting the expressions for each term:

$$q_x - q_{x+dx} - q_r = 0$$

Dividing by dx and taking the limit as $dx \rightarrow 0$, the difference $q_{x+dx} - q_x$ becomes a derivative:

$$\frac{dq_x}{dx} - \frac{q_r}{dx} = 0$$

Substituting Fourier's law $q_x = -kA \frac{dT}{dx}$:

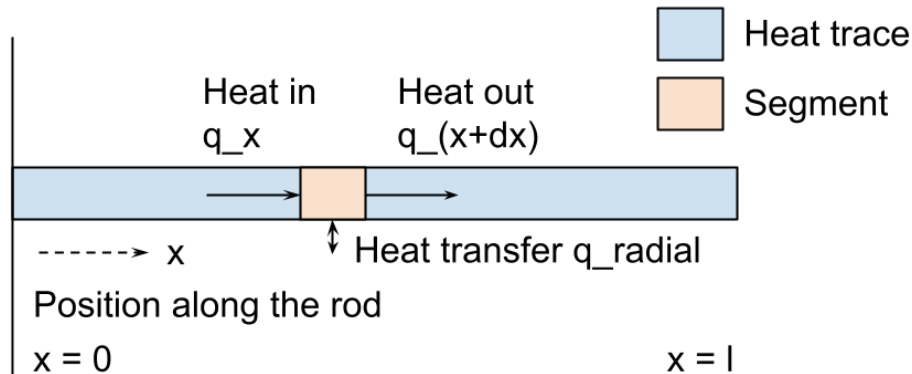
$$\frac{d}{dx} \left[-kA \frac{dT}{dx} \right] - \frac{q_r}{dx} = 0$$

Simplifying, assuming k and A are constant along the rod:

$$kA \frac{d^2 T}{dx^2} - \frac{q_r}{dx} = 0$$

This is the governing differential equation describing the temperature distribution along a rod exchanging heat with its surroundings. The terms represent:

- $kA \frac{d^2 T}{dx^2}$ - axial conduction into and out of the segment
- $\frac{q_r}{dx}$ - radial heat transfer with the surroundings $hP(T - T_\infty)$



It's convenient to express temperature relative to the surroundings by defining:

$$\theta = T - T_\infty$$

Substituting this gives:

$$\frac{d^2 \theta}{dx^2} - \frac{hP}{kA} \theta = 0$$

The grouped term

$$m^2 = \frac{hP}{kA}$$

is known as the square of the fin parameter. Using this definition, the governing equation takes the compact form:

$$\frac{d^2 \theta}{dx^2} - m^2 \theta = 0$$

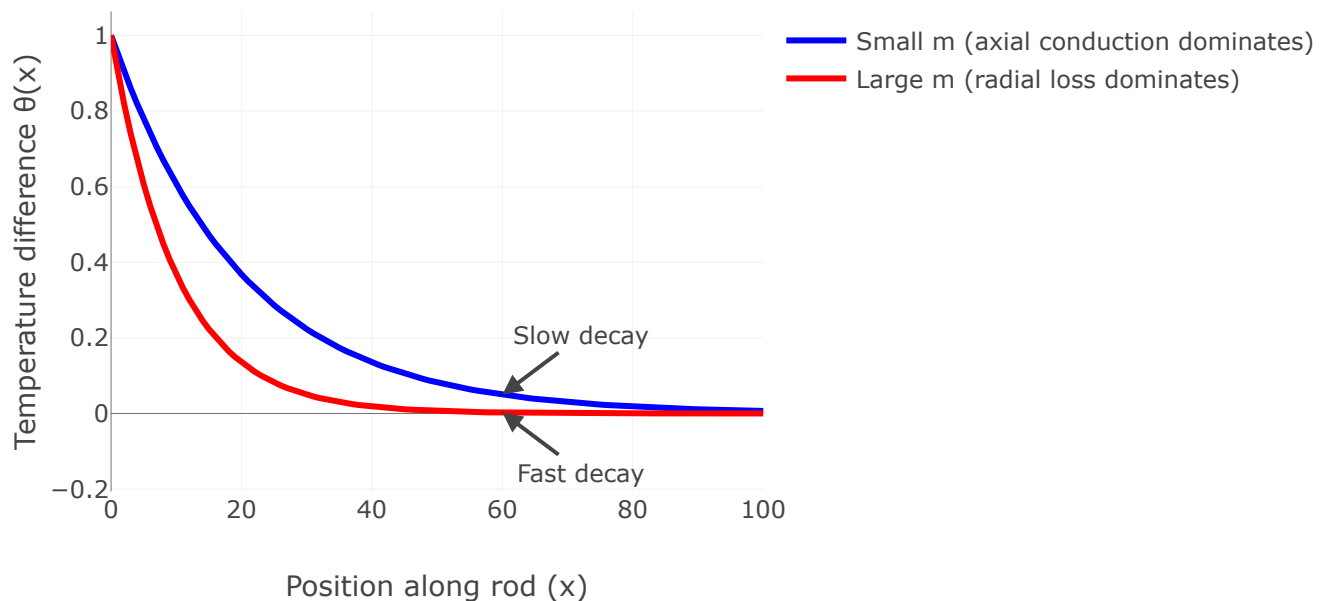
The fin parameter m characterises how strongly the rod exchanges heat with its surroundings relative to how effectively heat is conducted along its length.

- h - strength of heat transfer between the rod surface and the surrounding fluid
- P - perimeter of the rod influencing the heat exchange area
- k - how effectively heat can conduct along the rod material
- A - cross-sectional area for axial conduction

Physically, this equation describes the balance between axial conduction within the rod and heat exchange with the environment along its surface.

- If m is small, conduction along the rod dominates and temperature changes slowly with distance
- If m is large, heat exchange with the surroundings dominates and the temperature approaches the surrounding temperature rapidly

Governing Differential Equation: Effect of m on Temperature Decay



Why Exponentials Appear in the Solution

Now we have the governing differential equation for temperature along the rod, but why do exponential's appear in the solutions?

An exponential function $y = e^{rx}$ has the property that its derivative is proportional to itself:

$$\frac{d}{dx}e^{rx} = re^{rx}, \quad \frac{d^2}{dx^2}e^{rx} = r^2e^{rx}$$

If we take the governing differential equation $\frac{d^2\theta}{dx^2} - m^2\theta = 0$ and try a solution of a known form with an unknown parameter r :

$$\theta(x) = e^{rx}$$

and substitute it into the differential equation:

$$\begin{aligned} \frac{d^2\theta}{dx^2} - m^2\theta &= r^2e^{rx} - m^2e^{rx} = 0 \\ e^{rx}(r^2 - m^2) &= 0 \end{aligned}$$

Dividing both sides by e^{rx} :

$$r^2 - m^2 = 0 \quad \Rightarrow \quad r = \pm m$$

This confirms the exponential term we tried is a solution, and there are two variants that satisfy the differential equation:

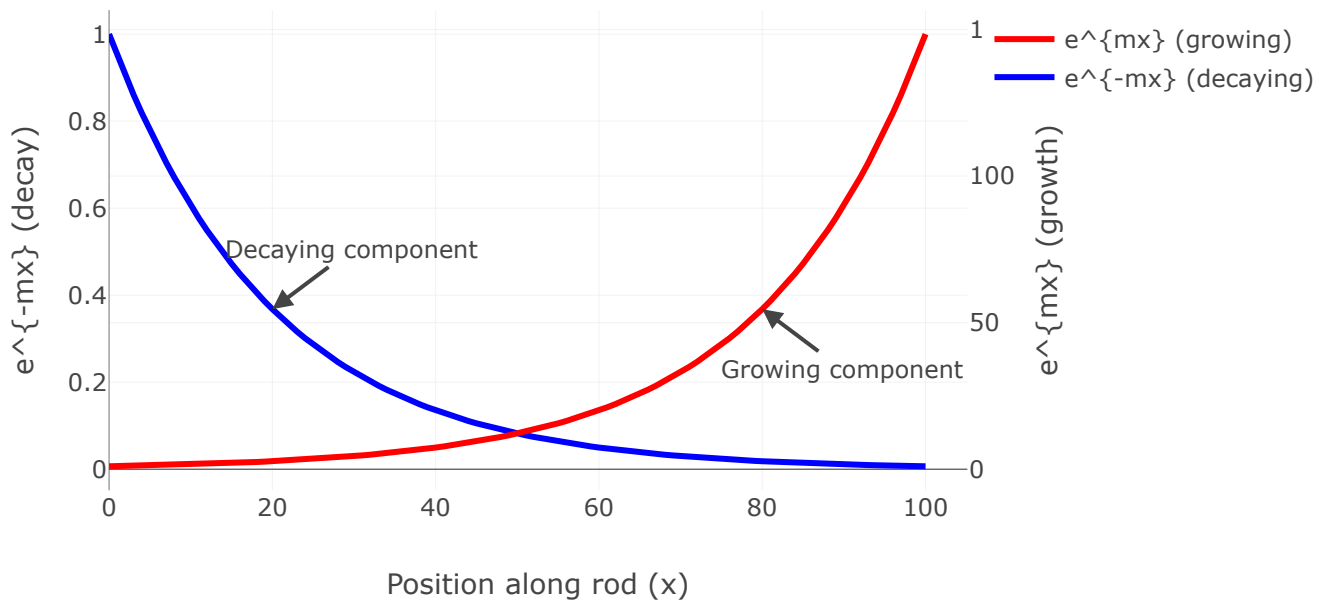
$$e^{mx} \quad \text{and} \quad e^{-mx}$$

Because the differential equation is linear (the unknown function θ and its derivatives appear only next to the first power) and homogeneous (the right-hand side is zero), any linear combination of valid solutions is also a solution. Therefore the general solution to the fin temperature equation is:

$$\theta(x) = C_1e^{mx} + C_2e^{-mx}$$

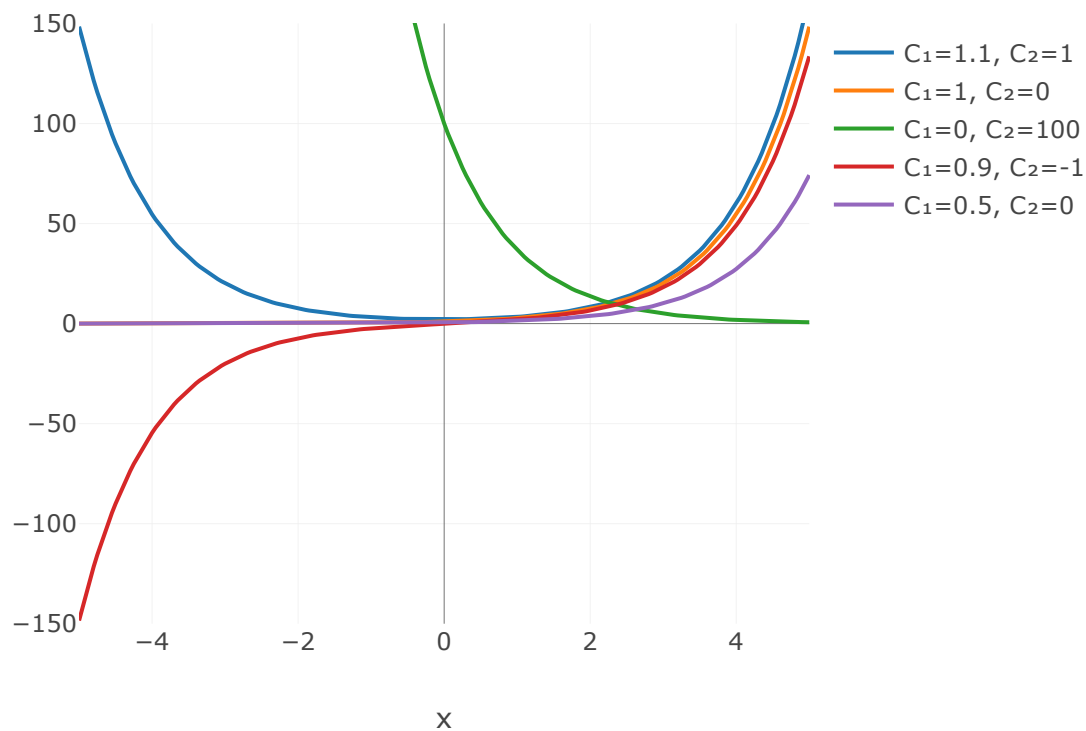
- C_1 and C_2 are constants determined by boundary conditions
- The term e^{-mx} decays along the rod; the term e^{mx} grows, which may or may not be physically relevant depending on boundary conditions

Exponential Components of $\theta(x)$



The figure shows how the two exponential components behave along the rod. One grows and one decays.

The plot below shows how changing the constants C_1 and C_2 changes the shape of the solution while still satisfying the differential equation.



Hyperbolic Functions

While the general solution to the governing differential equation $\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$ is mathematically correct, expressing it in a hyperbolic form can make applying boundary conditions easier. Hyperbolic sine and cosine are defined as linear combinations of exponentials:

$$\cosh(mx) = 1 + \frac{(mx)^2}{2!} + \frac{(mx)^4}{4!} + \dots$$

$$\sinh(mx) = mx + \frac{(mx)^3}{3!} + \frac{(mx)^5}{5!} + \dots$$

$$e^{mx} = 1 + mx + \frac{(mx)^2}{2!} + \frac{(mx)^3}{3!} + \frac{(mx)^4}{4!} + \dots$$

$$e^{-mx} = 1 - mx + \frac{(mx)^2}{2!} - \frac{(mx)^3}{3!} + \frac{(mx)^4}{4!} - \dots$$

$$\cosh(mx) = \frac{e^{mx} + e^{-mx}}{2}, \quad \sinh(mx) = \frac{e^{mx} - e^{-mx}}{2}$$

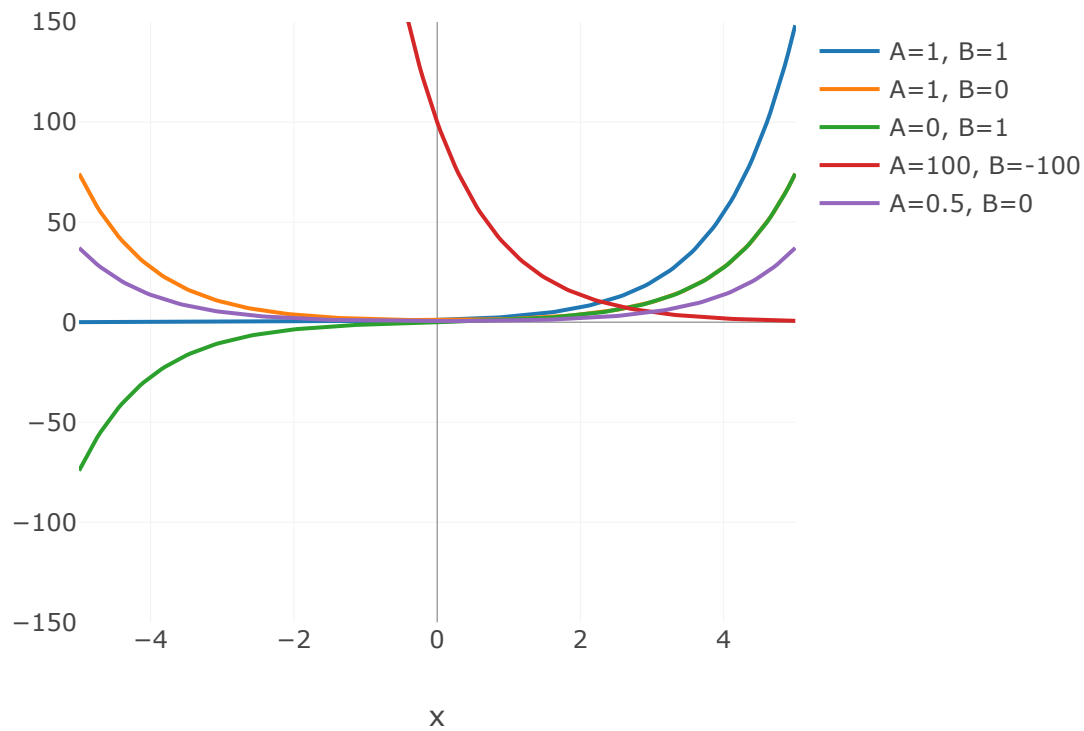
Rearrange:

$$e^{mx} = \cosh(mx) + \sinh(mx), \quad e^{-mx} = \cosh(mx) - \sinh(mx)$$

And the solution can be written as:

$$\theta(x) = A \cosh(mx) + B \sinh(mx)$$

- A and B are constants determined by boundary conditions (like fixed base temperature or insulated tip)
- This form is convenient because $\cosh$ is symmetric and $\sinh$ is antisymmetric about $x = 0$



By substituting $\cosh(mx) = \frac{e^{mx} + e^{-mx}}{2}$ and $\sinh(mx) = \frac{e^{mx} - e^{-mx}}{2}$ into the hyperbolic form of the solution, we can determine that the constants A and B relate to the exponential constants C_1 and C_2 as follows:

$$C_1 = \frac{A + B}{2}, \quad C_2 = \frac{A - B}{2}$$

Meaning that if $A = 1$ and $B = 1$ then $C_1 = 1$ and $C_2 = 0$.

Exponential vs Hyperbolic Forms

Rewriting the solution in terms of $\cosh$ and $\sinh$ allows us to easily apply boundary conditions and derive the final analytical formula. This is because hyperbolic functions behave very conveniently at the base:

$$\cosh(0) = 1, \quad \sinh(0) = 0$$

So if

$$\theta(x) = A \cosh(mx) + B \sinh(mx)$$

then at $x = 0$:

$$\theta(0) = A$$

The constant **A** directly equals the base temperature difference.

However with the exponential form of the differential equation $\theta(x) = C_1 e^{mx} + C_2 e^{-mx}$ the solution at $x = 0$ would be :

$$\theta(0) = C_1 + C_2$$

which is less convenient. But certain situations exist where the exponential form is cleaner, such as for an infinitely long fin, the boundary condition

$$\theta(x \rightarrow \infty) = 0$$

forces the growing exponential term e^{mx} to be zero (otherwise the temperature would diverge). This eliminates one constant:

$$C_1 = 0$$

leaving

$$\theta(x) = C_2 e^{-mx}$$

And the exponential form is more compact for this case.

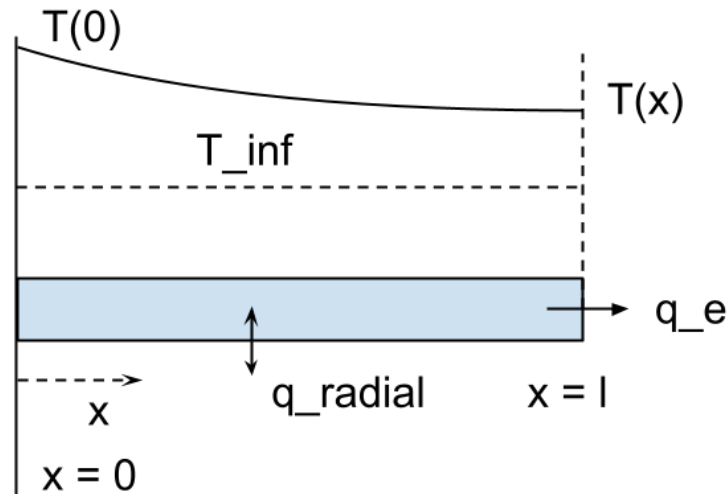
Boundary Conditions

To determine the temperature distribution along the rod we apply boundary conditions to the general solution of the governing differential equation:

$$\theta(x) = A \cosh(mx) + B \sinh(mx)$$

1. Thermally Constant Base and Convective Heat Transfer at the Tip

Base of the Rod ($x = 0$)



The rod is thermally anchored at the base, so its temperature difference relative to the surroundings is:

$$\theta(0) = \theta_b$$

Substituting $x = 0$ into the general solution:

$$\theta_b = A \cosh(0) + B \sinh(0)$$

Using the identities:

$$\cosh(0) = 1, \quad \sinh(0) = 0$$

we obtain:

$$\theta_b = A$$

So the temperature distribution becomes:

$$\theta(x) = \theta_b \cosh(mx) + B \sinh(mx)$$

Tip of the Rod ($x = L$)

At the tip of the rod, heat arriving by conduction along the rod must equal the heat leaving the tip. This heat loss can occur either by convection or conduction.

Heat arriving at the tip by conduction along the rod is given by Fourier's law:

$$q_x|_{x=L} = -k_f A_k \left(\frac{dT}{dx} \right)_{x=L}$$

- k_f - thermal conductivity of the rod
- A_k - cross-sectional area of the rod

Convective Heat Loss from the Tip

If the tip is exposed to a surrounding fluid, heat leaves the tip by convection:

$$q_e = h_e A_k (T_L - T_\infty)$$

At steady state, heat conducted to the tip equals heat leaving the tip:

$$-k_f A_k \left(\frac{dT}{dx} \right)_{x=L} = h_e A_k (T_L - T_\infty)$$

Dividing by $k_f A_k$:

$$-\left(\frac{dT}{dx} \right)_{x=L} = \frac{h_e}{k_f} (T_L - T_\infty)$$

Since $\theta = T - T_\infty$, this becomes:

$$-\left(\frac{d\theta}{dx} \right)_{x=L} = \frac{h_e}{k_f} \theta(L)$$

or equivalently:

$$\frac{d\theta}{dx} \Big|_{x=L} + \frac{h_e}{k_f} \theta(L) = 0$$

Applying the Boundary Condition to the Hyperbolic Solution

The general solution of the governing differential equation is

$$\theta(x) = A \cosh(mx) + B \sinh(mx)$$

Its derivative is

$$\frac{d\theta}{dx} = Am \sinh(mx) + Bm \cosh(mx)$$

Plugging these into the equation $\frac{d\theta}{dx} \Big|_{x=L} + \frac{h_e}{k_f} \theta(L) = 0$ for the boundary condition at $x = L$:

$$Am \sinh(mL) + Bm \cosh(mL) + \frac{h}{k} (A \cosh(mL) + B \sinh(mL)) = 0$$

Solving for B

Expand the expression:

$$Am \sinh(mL) + Bm \cosh(mL) + \frac{h}{k} A \cosh(mL) + \frac{h}{k} B \sinh(mL) = 0$$

Group the constants:

$$A \left[m \sinh(mL) + \frac{h}{k} \cosh(mL) \right] + B \left[m \cosh(mL) + \frac{h}{k} \sinh(mL) \right] = 0$$

Solving for B :

$$B = -A \frac{m \sinh(mL) + \frac{h}{k} \cosh(mL)}{m \cosh(mL) + \frac{h}{k} \sinh(mL)}$$

Using the base boundary condition $A = \theta_b$:

$$B = -\theta_b \frac{m \sinh(mL) + \frac{h}{k} \cosh(mL)}{m \cosh(mL) + \frac{h}{k} \sinh(mL)}$$

Substituting Back into the Solution

$$\theta(x) = \theta_b \cosh(mx) - \theta_b \frac{m \sinh(mL) + \frac{h}{k} \cosh(mL)}{m \cosh(mL) + \frac{h}{k} \sinh(mL)} \sinh(mx)$$

Dividing through by θ_b gives the normalised temperature distribution:

$$\frac{\theta(x)}{\theta_b} = \cosh(mx) - \frac{m \sinh(mL) + \frac{h}{k} \cosh(mL)}{m \cosh(mL) + \frac{h}{k} \sinh(mL)} \sinh(mx)$$

Using hyperbolic identities and rearranging terms, this expression can be written in a more compact form:

$$\frac{\theta(x)}{\theta_b} = \frac{\cosh(m(L-x)) + \frac{h}{mk} \sinh(m(L-x))}{\cosh(mL) + \frac{h}{mk} \sinh(mL)}$$

- θ_b - temperature difference at the base relative to the surroundings
- $\cosh(m(L-x))$ - describes the conductive temperature decay along the rod
- $\sinh(m(L-x))$ - accounts for the effect of heat loss from the tip
- $\frac{h}{mk}$ - dimensionless parameter representing the relative strength of tip convection

2. No Heat Transfer: Adiabatic (Insulated) Tip

If the tip is insulated, no heat leaves the end of the rod:

$$q_x|_{x=L} = 0$$

From Fourier's law this means the temperature gradient at the tip must be zero:

$$\left(\frac{dT}{dx}\right)_{x=L} = 0$$

Since $\theta = T - T_\infty$, the same condition applies to θ :

$$\left(\frac{d\theta}{dx}\right)_{x=L} = 0$$

Applying the Boundary Condition to the Hyperbolic Solution

The general solution for the temperature distribution along the rod is:

$$\theta(x) = A \cosh(mx) + B \sinh(mx)$$

The derivative of this expression is:

$$\frac{d\theta}{dx} = Am \sinh(mx) + Bm \cosh(mx)$$

Plugging these into the equation $\frac{d\theta}{dx}|_{x=L} = 0$ for the adiabatic boundary condition at $x = L$:

$$Am \sinh(mL) + Bm \cosh(mL) = 0$$

Solving for B

Divide through by m :

$$A \sinh(mL) + B \cosh(mL) = 0$$

Rearranging gives:

$$B \cosh(mL) = -A \sinh(mL)$$

so

$$B = -A \tanh(mL)$$

Using the base boundary condition $A = \theta_b$:

$$B = -\theta_b \tanh(mL)$$

Substituting Back into the Solution

Substituting A and B into the general solution:

$$\theta(x) = \theta_b \cosh(mx) - \theta_b \tanh(mL) \sinh(mx)$$

Dividing through by θ_b gives the normalised temperature distribution:

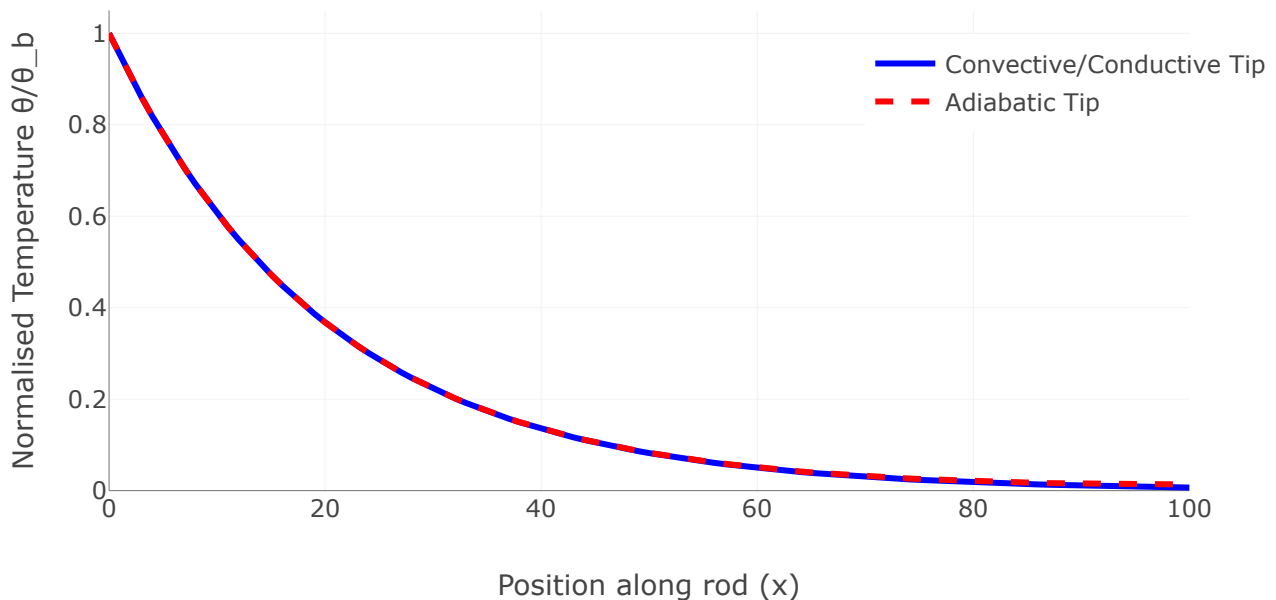
$$\frac{\theta(x)}{\theta_b} = \cosh(mx) - \tanh(mL) \sinh(mx)$$

Using hyperbolic identities, this expression can be rearranged into:

$$\frac{\theta(x)}{\theta_b} = \frac{\cosh(m(L-x))}{\cosh(mL)}$$

- θ_b - temperature difference at the base relative to surroundings
- $\cosh(m(L-x))$ - describes how temperature decays along the rod
- $\cosh(mL)$ - ensures the base temperature boundary condition is satisfied

Normalised Temperature Along Rod



Worked Example

Using the following values:

- $T_{\text{pipe}} = 30 \text{ K}$
- $T_{\text{base}} = 300 \text{ K}$
- $L = 0.4 \text{ m}$
- $r = 0.003 \text{ m}$
- $k_{\text{rod}} = 15 \text{ W}/(\text{m}\cdot\text{K})$
- $h = 5 \text{ W}/(\text{m}^2\cdot\text{K})$

Step 1: Define Temperature Difference

$$\theta(x) = T(x) - T_{\text{pipe}}$$

$$\theta_b = T_{\text{base}} - T_{\text{pipe}} = 300 - 30 = 270 \text{ K}$$

Step 2: Fin Parameter

$$m = \sqrt{\frac{hP}{kA}}$$

For a cylindrical rod:

$$P = 2\pi r, \quad A = \pi r^2, \quad m = \sqrt{\frac{2h}{kr}}$$

Substituting values:

$$m = \sqrt{\frac{2 \cdot 5}{15 \cdot 0.003}} = \sqrt{\frac{10}{0.045}} = \sqrt{222.2} \approx 14.9 \text{ m}^{-1}$$

Step 3: Temperature Distribution

Using the adiabatic tip solution:

$$\frac{\theta(x)}{\theta_b} = \frac{\cosh(m(L-x))}{\cosh(mL)}$$

Converting back to temperature:

$$T(x) = \frac{\cosh(m(L-x))}{\cosh(mL)}(T_{\text{base}} - T_{\text{pipe}}) + T_{\text{pipe}}$$

Step 4: Evaluate**At the base ($x = 0$):**

$$T(0) = 300 \text{ K}$$

At the tip ($x = L = 0.4$):

$$mL = 14.9 \cdot 0.4 \approx 5.96$$

$$\cosh(5.96) \approx 193$$

$$T(L) = \frac{1}{193} \cdot 270 + 30 \approx 31.4 \text{ K}$$

Electrical Analogies

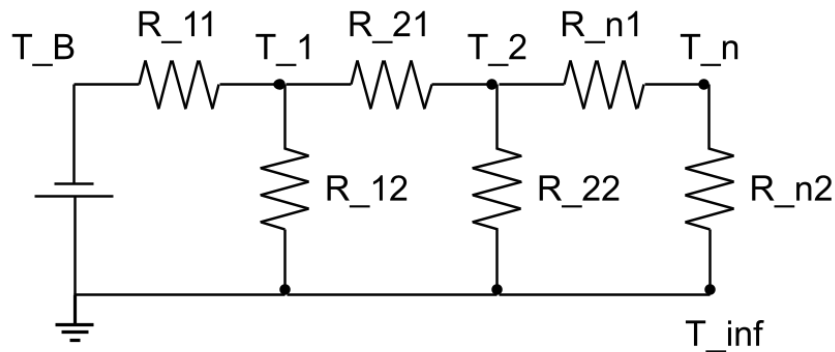
Another way to model heat transfer along a rod is to represent it as an electrical resistor network.

The equivalent physical properties are:

- Temperature $T \rightarrow$ Voltage V
- Heat flow $q \rightarrow$ Electrical current I
- Thermal resistance $R_{th} \rightarrow$ Electrical resistance R

Using this analogy, the rod can be divided into many small segments with each segment containing:

- An axial resistance representing conduction along the rod
- A radial resistance representing heat transfer to the surroundings



Thermal Resistances

For a rod segment of length dx , axial conduction resistance:

$$R_R = \frac{dx}{kA}$$

- k - thermal conductivity of the rod
- A - cross-sectional area

And radial heat transfer resistance with the surroundings:

$$R_H = \frac{1}{hP dx}$$

- h - heat transfer coefficient
- P - perimeter of the rod

In both cases, thermal resistances R_R and R_H have units of $\frac{K}{W}$, representing the temperature difference required to drive a unit of heat flow.

Node Energy Balance

At each node energy conservation must be satisfied.

First we consider a simple case with only two thermal resistances connected to the node. Heat arrives from the left through resistance R_1 and leaves to the surroundings through resistance R_2

Using the thermal equivalent of Ohm's law:

$$q = \frac{\Delta T}{R}$$

The heat flows into and out of the node are:

$$q_1 = \frac{T_{i-1} - T_i}{R_1}$$
$$q_2 = \frac{T_\infty - T_i}{R_2}$$

At steady state, conservation of energy requires:

$$q_1 + q_2 = 0$$

Substituting the expressions for the heat flows gives:

$$\frac{T_{i-1} - T_i}{R_1} + \frac{T_\infty - T_i}{R_2} = 0$$

Rearranging for the node temperature:

$$T_i = \frac{\frac{T_{i-1}}{R_1} + \frac{T_\infty}{R_2}}{\frac{1}{R_1} + \frac{1}{R_2}}$$

This is the same algebra as when solving a simple electrical node with two resistors.

The full ladder network adds a third heat path to the same node representing conduction from the right neighbour:

$$\frac{T_{i-1} - T_i}{R_R} + \frac{T_{i+1} - T_i}{R_R} + \frac{T_\infty - T_i}{R_H} = 0$$

Rearranging for node temperature:

$$T_i = \frac{\frac{T_{i-1}}{R_R} + \frac{T_{i+1}}{R_R} + \frac{T_\infty}{R_H}}{\frac{2}{R_R} + \frac{1}{R_H}}$$

This equation is identical to applying Kirchhoff's Current Law in the electrical circuit analogue.

Connection to the Fin Equation

The ladder network becomes equivalent to the continuous fin equation when the segment size becomes very small.

If we substitute

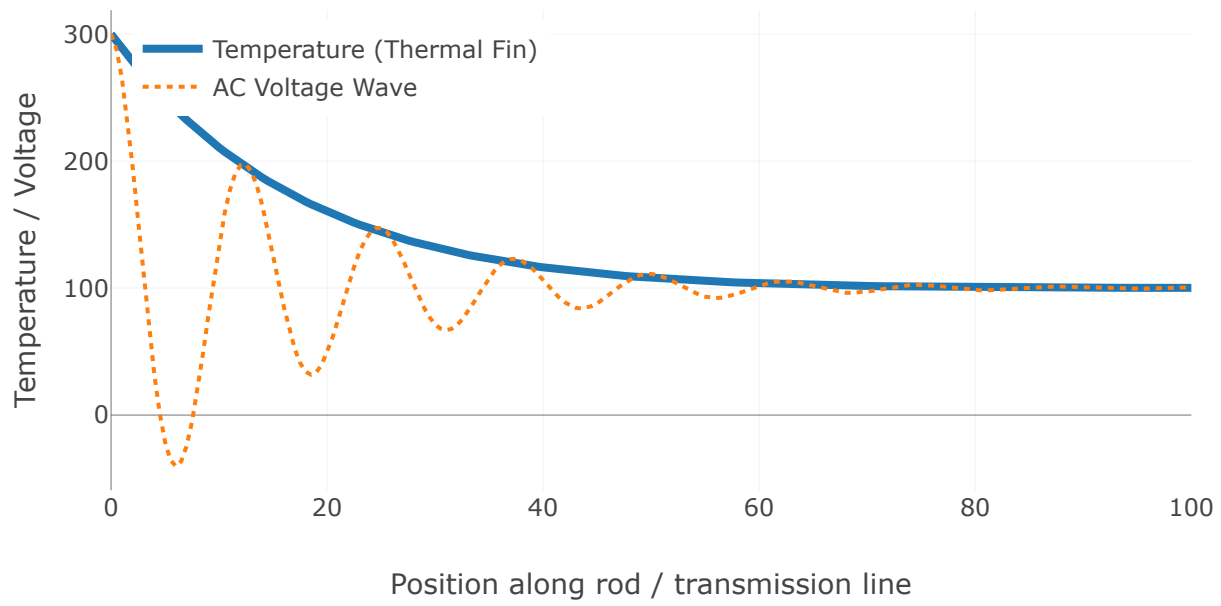
$$R_R = \frac{dx}{kA} \quad \text{and} \quad R_H = \frac{1}{hP dx}$$

into the nodal energy balance, and the excess temperature substitution $\theta = T - T_\infty$ is applied, as the segment length $dx \rightarrow 0$, the discrete ladder model converges to the continuous fin equation:

$$\frac{d^2\theta}{dx^2} - m^2\theta = 0$$

The resistor ladder is a finite-difference approximation of the governing heat conduction equation.

Thermal Fin vs Attenuating Electrical Wave



Including Thermal Capacitance and Time Response

To model transient behaviour each node must store thermal energy, achieved by adding a thermal capacitance at each node:

$$C = mc$$

- m - mass of the segment
- c - specific heat capacity

The node energy balance now becomes:

$$\frac{T_{i-1} - T_i}{R_R} + \frac{T_{i+1} - T_i}{R_R} + \frac{T_\infty - T_i}{R_H} = C \frac{dT_i}{dt}$$

This equation describes how temperature evolves over time due to both heat flow and stored energy. It is identical to the equation we use for the finite element method in the section below.

Time Lag and Dynamic Response

The addition of capacitance introduces a time lag in the system response. Instead of temperatures adjusting instantly, each node responds gradually depending on its thermal time constant:

$$\tau = RC$$

This is directly analogous to an electrical RC circuit, where voltage lags behind current.

Physically, this means:

- Large thermal mass $\rightarrow$ slower response
- High resistance $\rightarrow$ slower heat transfer $\rightarrow$ slower response

Role of Imaginary Numbers (Frequency Domain View)

If the system is analysed in the frequency domain (for example under oscillating heat input), the time derivative becomes:

$$\frac{d}{dt} \rightarrow i\omega$$

and the governing equation becomes:

$$\frac{T_{i-1} - T_i}{R_R} + \frac{T_{i+1} - T_i}{R_R} + \frac{T_\infty - T_i}{R_H} = i\omega C T_i$$

Here, the capacitance introduces a phase lag between heat input and temperature response, just like an electrical capacitor. The imaginary term $i\omega C$ represents energy storage and delayed response.

Lumped Capacitance Simplification

If the temperature within the body can be assumed uniform, the entire system can be reduced to a single node with one capacitance:

$$C \frac{dT}{dt} = Q$$

which has the simple solution:

$$T(t) = T_{ss} + (T_0 - T_{ss})e^{-t/\tau}$$

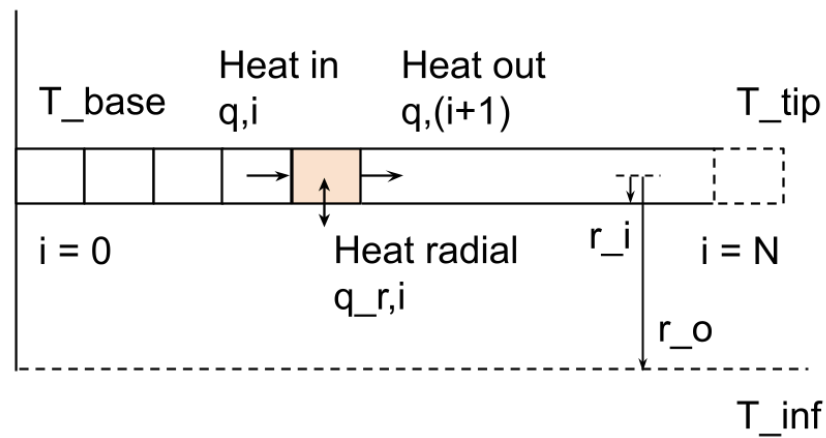
where T_{ss} is the steady-state temperature and τ is the time constant.

$$T_{ss} = \frac{\frac{T_0}{R_H} + \frac{T_\infty}{R_R}}{\frac{1}{R_H} + \frac{1}{R_R}}$$

This avoids spatial discretisation entirely and provides a closed-form solution without iteration.

Finite Element Method

Another way to model heat transfer along the rod is the finite element method (FEM). This approach can simulate the transient thermal behaviour of the rod as it evolves with time.



Each finite element represents a small rod segment of length dx with cross-sectional area A .

The element volume is:

$$V = A dx$$

and the thermal mass of the element is:

$$m = \rho V$$

The temperature of the element changes depending on the net heat flow into the element. Applying conservation of energy over a time step dt gives:

$$mc \frac{dT}{dt} = q_{left} + q_{right} - q_{radial}$$

where

- c - specific heat capacity of the rod material

and the three heat flows correspond to:

- Axial conduction from the neighbouring element on the left
- Axial conduction from the neighbouring element on the right
- Radial heat transfer between the rod and the surroundings

Axial Conduction

Heat transfer between neighbouring elements is calculated using Fourier's law of conduction:

$$q = kA \frac{\Delta T}{dx}$$

This expression describes the heat flowing between adjacent elements due to the temperature difference along the rod.

Radial Heat Transfer

Heat can also exchange radially with the surroundings. This radial conduction through the cylindrical gap can be described using the logarithmic resistance of a cylindrical wall:

$$q_{radial} = \frac{2\pi k dx (T - T_{\infty})}{\ln(r_o/r_i)}$$

- k - thermal conductivity of the surrounding material
- r_i - rod radius
- r_o - outer radius of the surrounding region

This expression represents heat conduction through the cylindrical layer surrounding the rod.

Transient Temperature Update

The element temperature is updated using the energy balance:

$$\Delta T = \frac{q_{net} dt}{mc}$$

where

$$q_{net} = q_{left} + q_{right} - q_{radial}$$

By repeatedly applying this update over many small time steps, the temperature distribution gradually evolves until it reaches steady state.

Boundary Conditions

Consider the base of the rod has a fixed temperature:

$$T(0) = T_{base}$$

As the simulation progresses, heat flows from the base along the rod and into the surroundings until a stable distribution is reached.

Relationship to the Fin Equation

As the element size becomes small and the time step approaches zero, the transient finite element model approaches the same governing equation used in the analytical solution:

$$\frac{d^2\theta}{dx^2} - m^2\theta = 0$$

The finite element model provides a transient solution of the same physical problem.

Conclusion

We have demonstrated the derivation of the analytical temperature distribution along a cylindrical rod with radial heat transfer.

- Energy conservation and Fourier's law form the foundation for the governing differential equation
- Small rod segments allow us to quantify axial conduction and radial heat transfer
- Second-order differential equations naturally produce exponential solutions, which can be recast as hyperbolic functions for easier application of boundary conditions
- Boundary conditions determine the constants in the solution, leading to the final analytical expression for $\theta(x)$.
- The parameter $m = \sqrt{hP/kA}$ controls the rate of temperature decay along the rod